MTH2021 Mathematical Methods 2

Self-study sheet 5 – solutions

1.
(a) The function

$$f(z) = \frac{1+z^2}{z(z-2)^2} \tag{1}$$

has a simple pole (a pole of order 1) at $z = 0$ and a double pole (a pole of order 2) at $z = 2$. We can use the result

$$r = \frac{1}{(m-1)!} \lim_{z \to z_0} \frac{d^{m-1}}{dz^{m-1}} \left[(z-z_0)^m f(z)\right] \tag{2}$$

for the residue, r, at a pole of order m at z_0.

At $z_0 = 0$ we have a pole of order $m = 1$ and the residue is

$$r = \lim_{z \to 0} (z f(z)) = \lim_{z \to 0} \left(\frac{1+z^2}{(z-2)^2}\right) = \frac{1}{4} \tag{3}$$

At $z_0 = 2$ we have a pole of order $m = 2$ and the residue is

$$\begin{aligned} r &= \frac{1}{1!} \lim_{z \to 2} \frac{d}{dz} \left[(z-2)^2 f(z)\right] && (4) \\ &= \lim_{z \to 2} \frac{d}{dz} \left(\frac{1+z^2}{z}\right) && (5) \\ &= \lim_{z \to 2} \left(-\frac{1}{z^2} + 1\right) = \frac{3}{4} && (6) \end{aligned}$$

(b) The function

$$f(z) = \frac{\exp z}{(z-\mathrm{i})^3} \tag{7}$$

has a pole of order $m = 3$ at $z_0 = \mathrm{i}$. The residue is

$$\begin{aligned} r &= \frac{1}{2!} \lim_{z \to \mathrm{i}} \frac{d^2}{dz^2} \left[(z-\mathrm{i})^3 f(z)\right] && (8) \\ &= \frac{1}{2!} \lim_{z \to \mathrm{i}} \frac{d^2}{dz^2} \exp z && (9) \\ &= \frac{1}{2!} \lim_{z \to \mathrm{i}} \exp z && (10) \\ &= \frac{\cos 1 + \mathrm{i} \sin 1}{2} && (11) \end{aligned}$$

(c) For the function

$$f(z) = \frac{1 - \cos z}{z^5} \tag{12}$$

it's easiest to write down the Laurent series about $z = 0$ and read off the residue directly from there. Using the Taylor series for $\cos z$:

$$\begin{aligned} \frac{1-\cos z}{z^5} &= \frac{1-(1-z^2/2!+z^4/4!\ ...)}{z^5} && (13) \\ &= \frac{1}{2!}\frac{1}{z^3} - \frac{1}{4!}\frac{1}{z}\ ... && (14) \end{aligned}$$

and the residue at $z = 0$ (i.e. the coefficient of $1/z$) is $-1/4! = -1/24$.

2.

(a) We consider the integral

$$\int_{-\infty}^{\infty} \frac{x\exp(\mathrm{i}x)}{(a^2+x^2)(b^2+x^2)}\,dx \tag{15}$$

It is of the form

$$I = \int_{-\infty}^{\infty} \exp(\mathrm{i}\lambda x)\,\frac{P(x)}{Q(x)}\,dx \tag{16}$$

where $P(x)$ and $Q(x)$ are polynomials, such that:

- $Q(x)$ does not vanish for any real x
- the degree of $Q(x)$ is at least 1 greater than the degree of $P(x)$
- $\lambda > 0$.

(In this example $P(x) = x$, $Q(x) = (a^2 + x^2)(b^2 + x^2)$, $\lambda = 1$.)

From lectures

$$\int_{-\infty}^{\infty} \exp(\mathrm{i}\lambda x)\,\frac{P(x)}{Q(x)}\,dx = 2\pi\mathrm{i}\sum\left(\text{residues of } \exp(\mathrm{i}\lambda z)\,\frac{P(z)}{Q(z)} \text{ in upper half plane}\right) \tag{17}$$

In our case

$$\begin{aligned} \exp(\mathrm{i}\lambda z)\,\frac{P(z)}{Q(z)} &= \frac{z\exp(\mathrm{i}z)}{(z^2+a^2)(z^2+b^2)} && (18) \\ &= \frac{z\exp(\mathrm{i}z)}{(z-\mathrm{i}a)(z+\mathrm{i}a)(z-\mathrm{i}b)(z+\mathrm{i}b)} && (19) \end{aligned}$$

has simple poles at $z = \pm\mathrm{i}a$ and $z = \pm\mathrm{i}b$. Only two – at $\mathrm{i}a$ and $\mathrm{i}b$ – are in the upper half plane. The residue at $z = \mathrm{i}a$ is

$$\begin{aligned} r_1 &= \lim_{z\to\mathrm{i}a} \frac{(z-\mathrm{i}a)z\exp(\mathrm{i}z)}{(z-\mathrm{i}a)(z+\mathrm{i}a)(z-\mathrm{i}b)(z+\mathrm{i}b)} && (20) \\ &= -\frac{\exp(-a)}{2(a^2-b^2)} && (21) \end{aligned}$$

Similarly, the residue at $z = \mathrm{i}b$ is

$$r_2 = -\frac{\exp(-b)}{2(b^2-a^2)} \tag{22}$$

Therefore

$$\int_{-\infty}^{\infty} \frac{x\exp(\mathrm{i}x)}{(a^2+x^2)(b^2+x^2)}\,dx = 2\pi\mathrm{i}(r_1+r_2) = \pi\mathrm{i}\,\frac{\exp(-b)-\exp(-a)}{a^2-b^2} \tag{23}$$

Taking imaginary parts of both sides:

$$\int_{-\infty}^{\infty} \frac{x \sin x}{(a^2+x^2)(b^2+x^2)}\,dx = \pi\,\frac{\exp(-b)-\exp(-a)}{a^2-b^2} \tag{24}$$

(b) We consider the integral

$$\int_{-\infty}^{\infty} \frac{\exp(\mathrm{i}ax)}{(1+x^2)^2}\,dx \tag{25}$$

It is of the form

$$I = \int_{-\infty}^{\infty} \exp(\mathrm{i}\lambda x)\,\frac{P(x)}{Q(x)}\,dx \tag{26}$$

where $P(x)$ and $Q(x)$ are polynomials, such that:
- $Q(x)$ does not vanish for any real x
- the degree of $Q(x)$ is at least 1 greater than the degree of $P(x)$
- $\lambda > 0$.

(In this example $P(x) = 1$, $Q(x) = (1+x^2)^2$, $\lambda = a$.)

From lectures

$$\int_{-\infty}^{\infty} \exp(\mathrm{i}\lambda x)\,\frac{P(x)}{Q(x)}\,dx = 2\pi\mathrm{i}\sum\left(\text{residues of } \exp(\mathrm{i}\lambda z)\,\frac{P(z)}{Q(z)} \text{ in upper half plane}\right)$$

In our case

$$\begin{aligned} \exp(\mathrm{i}\lambda z)\,\frac{P(z)}{Q(z)} &= \frac{\exp(\mathrm{i}az)}{(1+z^2)^2} && (27)\\ &= \frac{\exp(\mathrm{i}az)}{(z-\mathrm{i})^2(z+\mathrm{i})^2} && (28) \end{aligned}$$

has poles of order 2 at $z = \pm\mathrm{i}$. Only one – at i – is in the upper half plane. The residue at $z = \mathrm{i}$ is

$$r = \frac{1}{1!}\lim_{z\to\mathrm{i}}\frac{d}{dz}\left[\frac{(z-\mathrm{i})^2\exp(\mathrm{i}az)}{(z-\mathrm{i})^2(z+\mathrm{i})^2}\right] = -\frac{\mathrm{i}(1+a)\exp(-a)}{4} \tag{29}$$

Therefore

$$\int_{-\infty}^{\infty} \frac{\exp(\mathrm{i}ax)}{(1+x^2)^2}\,dx = 2\pi\mathrm{i}r = \frac{\pi}{2}\,(1+a)\,\exp(-a) \tag{30}$$

Taking real parts:

$$\int_{-\infty}^{\infty} \frac{\cos(ax)}{(1+x^2)^2}\,dx = \frac{\pi}{2}\,(1+a)\,\exp(-a) \tag{31}$$

(c) The integral

$$\int_{-\infty}^{\infty} \frac{1+x^2}{1+x^4}\,dx \tag{32}$$

is of the form

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)}\,dx \tag{33}$$

where $P(x)$ and $Q(x)$ are polynomials such that:
• $Q(x)$ does not vanish for any real x
• the degree of $Q(x)$ is at least 2 greater than that of $P(x)$.
(In this example $P(x) = 1 + x^2$, $Q(x) = 1 + x^4$.)

From lectures

$$\int_{-\infty}^{\infty} \frac{P(x)}{Q(x)}\, dx = 2\pi \mathrm{i} \sum \left(\text{residues of } \frac{P(z)}{Q(z)} \text{ in upper half plane}\right) \tag{34}$$

The poles of the integrand occur whenever $z^4 + 1 = 0$, that is, at the four fourth roots of -1:

$$z_1 = \exp(\mathrm{i}\pi/4) = \frac{1+\mathrm{i}}{\sqrt{2}} \quad , \quad z_2 = \exp(3\pi\mathrm{i}/4) = \frac{-1+\mathrm{i}}{\sqrt{2}} \tag{35}$$

$$z_3 = \exp(5\pi\mathrm{i}/4) = \frac{-1-\mathrm{i}}{\sqrt{2}} \quad , \quad z_4 = \exp(7\pi\mathrm{i}/4) = \frac{1-\mathrm{i}}{\sqrt{2}} \tag{36}$$

Of these poles, only two – those at z_1 and z_2 – lie in the upper half plane. Both are simple poles (poles of order 1). To find the residues we can write

$$f(z) = \frac{P(z)}{Q(z)} = \frac{1+z^2}{1+z^4} = \frac{1+z^2}{(z-z_1)(z-z_2)(z-z_3)(z-z_4)} \tag{37}$$

The residue at z_1 is

$$r_1 = \lim_{z \to z_1} \left((z - z_1)\, f(z)\right) \tag{38}$$

$$= \lim_{z \to z_1} \left(\frac{1+z^2}{(z-z_2)(z-z_3)(z-z_4)}\right) \tag{39}$$

$$= \frac{1+z_1^2}{(z_1-z_2)(z_1-z_3)(z_1-z_4)} = \frac{1}{2\mathrm{i}\sqrt{2}} \tag{40}$$

Similarly, the residue at z_2 is

$$r_2 = \frac{1}{2\mathrm{i}\sqrt{2}} \tag{41}$$

Thus,

$$\int_{-\infty}^{\infty} \frac{1+x^2}{1+x^4}\, dx = 2\pi\mathrm{i}(r_1 + r_2) = \pi\sqrt{2} \tag{42}$$

3.
(a) By the Darboux inequality,

$$\left| \int_{C_R} \frac{z^2}{(1+z^2)^2}\, dz \right| \leq ML \tag{43}$$

where $L = \pi R$ is the length of the semicircle C_R and

$$M = \text{maximum of } \left| \frac{z^2}{(1+z^2)^2} \right| \text{ on } C_R \tag{44}$$

On C_R $z = R\exp(\mathrm{i}\theta)$ with $0 \leq \theta \leq \pi$. Thus

$$|z^2| = R^2 \tag{45}$$

Next consider

$$\begin{aligned} \left|(1+z^2)^2\right| &= \left|1+z^2\right|^2 && (46)\\ &= (1+z^2)(1+z^2)^* && (47)\\ &= [1+R^2\exp(2\mathrm{i}\theta)][1+R^2\exp(-2\mathrm{i}\theta)] && (48)\\ &= 1+2R^2\cos(2\theta)+R^4 \geq (R^2-1)^2 && (49) \end{aligned}$$

Thus on C_R

$$\left|\frac{z^2}{(1+z^2)^2}\right| \leq \frac{R^2}{(R^2-1)^2} \tag{50}$$

and hence

$$\left|\int_{C_R} \frac{z^2}{(1+z^2)^2}\,dz\right| \leq \frac{\pi R^3}{(R^2-1)^2} \tag{51}$$

which dies out as $R \to \infty$.

(b) We can sometimes use known results to save effort, e.g.

$$\int_{-\infty}^{\infty} \frac{x^2}{(1+x^2)^2}\,dx = \int_{-\infty}^{\infty} \frac{dx}{1+x^2} - \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^2} \tag{52}$$

The first integral on the right is π (we do not need contour integration for that) and the second integral is the limit of Question 2(b) as $a \to 0$, namely $\pi/2$:

$$\int_{-\infty}^{\infty} \frac{x^2}{(1+x^2)^2}\,dx = \pi/2 \tag{53}$$

We can do the last integral also by integration by parts.

4.
We have

$$z = \exp(\mathrm{i}\theta) \quad \Rightarrow \quad dz = \mathrm{i}\exp(\mathrm{i}\theta)d\theta \quad \Rightarrow \quad d\theta = \frac{dz}{\mathrm{i}z} \tag{54}$$

$$\cos\theta = (z+1/z)/2 \tag{55}$$

For $0 \leq \theta \leq 2\pi$, the complex variable $z = \exp(\mathrm{i}\theta)$ runs around the unit circle $|z| = 1$ once. Therefore

$$\begin{aligned} \int_0^{2\pi} \frac{d\theta}{5-3\cos\theta} &= \oint_{|z|=1} \frac{(dz/\mathrm{i}z)}{5-3(z+1/z)/2} && (56)\\ &= -\frac{2}{3\mathrm{i}}\oint_{|z|=1} \frac{dz}{(z-1/3)(z-3)} && (57) \end{aligned}$$

There is only one pole inside the circle $|z| = 1$, namely the simple pole at $z = 1/3$. The residue of

$$\frac{1}{(z-1/3)(z-3)} \tag{58}$$

at $z = 1/3$ is $1/(1/3 - 3) = -3/8$ and the integral is

$$\int_0^{2\pi} \frac{d\theta}{5 - 3\cos\theta} = -\frac{2}{3\mathrm{i}}\, 2\pi\mathrm{i}\,(-3/8) = \pi/2 \tag{59}$$